

Enabling Test Case Prioritization For Component Based Software Development

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Abstract— Frequent evolution of modern software systems increases complexity and system failure. Therefore, to overcome these issues the component-based system was developed that consists of integrated reuse components that work together to perform specific tasks or new application development. Mostly component based systems use configuration capabilities to adopt the changes and external uncertainties. Quality of these systems can be assessed through customer satisfaction after modification to verify the performance during maintenance activities. Software engineers for high-reliability, overall requirements functionality needs to verify before release of new product. One of the common ways to evaluate system quality in a sequence of releases is regression testing. Software quality engineers use to ensure that no faults introduced after changes. The purpose of this research work to identify the limitations in existing studies that create complexity and increase efforts during testing component based software and eliminate those factors through the proposed approach. The mixed research methodology will be followed in this context comprising qualitative and quantitative methods. To investigate the potential benefits of the proposed approach; will be performed a case study and experiment. Preliminary results indicated that the proposed approach significantly improve faults detection rate after changes alongwith less effort and cost in component based system development.

Keywords— *Component Based Software, Regression Testing, Test Case Prioritization*

I. INTRODUCTION

In many industrial setups today, software applications are considered as the heart and provided value to a great extent. Some of these industries are automotive, medical systems, process control, and manufacturing industries. Furthermore, for larger and complex software organization adopted Component-Based Software Engineering (CBSE) using Component-Based Development (CBD) approach to reduce complexity and making their delivery more agile leading to higher productivity [1], [2].

Thus, due to dynamic and configurable behavior evolve in uncertain environments. Hence, continuous changes lead to mishaps like system collapse, faults, low quality and complication in software maintenances in case of CBD [1]. Therefore, the quality of Component Based Software (CBS) is sensitive with the evolvement of changes and increase reliability risks due to new and old components testing [3]–[5]. During testing, verify that all the changes or modifications do not harm components behavior and system reliability. In [6] authors have defined that the process of verifying the customized software in the maintenance phase is commonly known as regression testing (RT) [7], [8]. RT is basically the re-execution of several subsets of tests that have been previously conducted [9]–[11]. In RT mostly re-execute all test cases after modification in the current version of

application. Whereas, in CBS as a different combination of components integration involved to build a single system, which increases the number of tests after every modification. Therefore, it is not a practical approach and ineffective to re-execute every test for each program function if once change occurs [9]. In the context of CBS, RT becomes even more expensive as each test needs to be executed against several different components [3], [11]. As a consequence of automating changes in requirements analysis, development and testing processes are needed that demand combined efforts from research and industry towards a broadly accepted solution [4], [5].

A. Problem Statement

Component based development has several limitations concerning its features modification and test case prioritization (TCP) of variabilities and commonalities after integration of components [1], [12]–[14] due to missing variabilities and frequently change test case historical information [6], [12]. The key issues are finding a way to formulate a component based development approach to facilitate new practitioners. Therefore, the aim of the study is firstly, to find and explore CBS development methods for software testing in current literature. Secondly, provide a solution to improve test case prioritization of CBS development after modification, which resolve the problems of the industrial practitioner.

II. RELATED WORK

In component based system during changes customer perspective ignored and changes handle using previous analysis about components while reusing and updating components. For the user-centric model during components modification and development [5] proposed a model which completely based on customer participation, feedback, and analysis. Through the questioner model, they compared existing models and results depict that new model outperformed due to active feedback of customers. Thus, it is identified that customer involvement also important during risk management, change management, and system maintainability.

In [6] suggested that reuse components verification is an important activity from requirement analysis to system design using test case execution for system reliability. The evaluation results validate that system response normal when requirements based coverage analysis of reuse components verification as metric after modifications. As the size of the system grow and more functionalities added than a chance of maintainability increases. To avoid the chance of maintainability and more test execution in mobile app testing after changes are needed.

Therefore, [15] proposed automated test case selection and prioritization technique by semantically indexing test cases and select according to each change request. Thus, evaluated that AutoTestPlan approach can identify desirable test cases. Hence, there is need to identified relation amongst requirement change to test cases change frequency and source code change to test cases textual relation.

Additionally, for changes impact analysis [10] proposed similar requirements or coverage based clustering for system verification to prioritize test cases. While authors [3] discussed that system reliability also important in configurable systems. Therefore, they proposed RT technique based on time criteria during the evolutionary cycle to verify modifications. In [1] proposed techniques for component based software for system verification from components specification to test case analysis. Evaluation results showed that test case generation improves with components specification for test case analysis. In case of component based applications test cases of dependent components functionality may need prioritization process for verification and modified test case criteria.

In [9] discussed that code based approach not significant to verify system reliability in case of large and complex CBS. To overcome existing issues authors proposed a model-based approach based on dependencies among components by mining historical data for system performance evaluation. Therefore, during the maintainability process of components for RT, have to consider components', change information, for better reliability and productivity using different data mining techniques.

From the literature review, we can analyze that large and complex software applications are divided into small

activities. In addition, most of these applications constructed reusing different parts or components to implement modification within limited resources. Hence, reliability issue initiated when there is a change in system components [6], [10]-[11], [17], [19] and complex to deal components testing in maintainability phase [6], [16-19] due to a large set of software components and their test cases. Thus, need a technique which deals to verify that the performance of components does not affect after changes and increase the fault detection rate within cost and time.

A. Research Questions

To achieve our objectives, we use the following research questions:

RQ 1: What are the key practices which can be useful to minimize gaps from state of art practice for CBS testing?

RQ 2: What could characterize a proposed approach for CBS testing?

RQ 3: Can we identify key components be combined to construct a tool to support and enhance the proposed approach?

RQ 4: Does method and tool combine improve the faults detection rates?

III. METHODOLOGY

The descriptive research will be done in which stories are reported. Through descriptive research, we have identified our domain, identified relevant gaps and challenges occurred. The research method as shown in Figure 1 for conducting this research starts with a mixed method comprising both qualitative and quantitative.

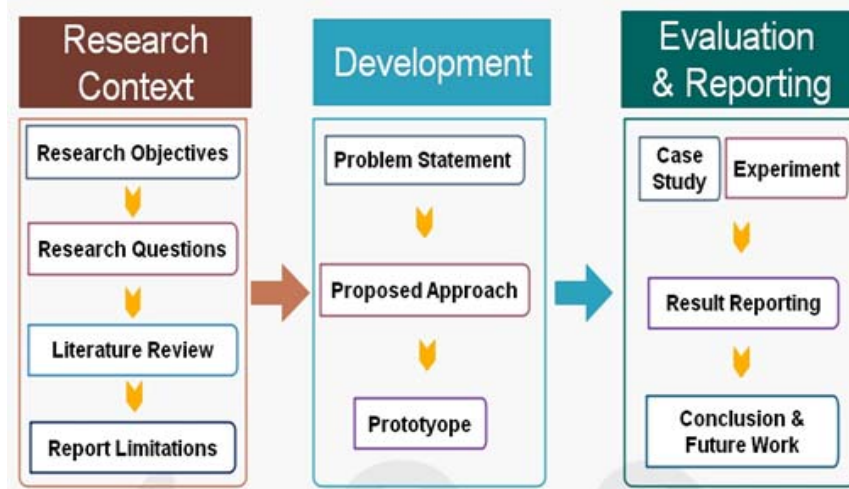


Figure 1. Research overview

The methodology divides into three phases i.e. research context, formulation of problem statement and evaluation. In the research context, we review existing literature to achieve define objectives and research questions to report issues and challenges. Then formulate the problem statement and conceptual framework steps defined to develop the proposed framework.

In the evaluation phase, we define three methods of evaluation i.e. experiment, case study, and implementation

that will be used to investigate the proposed approach (PA). Then on these methods, we will be report results and then conclude the work with the future direction of our work. To answer research question 1 and 2, analyzed the existing literature to identify the trends, opportunities, and challenges in CBS for higher productivity and user satisfaction to formulate PA. Which will then be empirically evaluated through industrial evaluation

A. Proposed Approach

In this research, proposed a method for test case prioritization of component based system development. The PA will provide, a solution to mitigate problems formulated after literature review (possible proposed approach is shown in Figure 2).

The PA base on the principles and practices of CBS development for software testing. After studying and caring analysis of the selected papers the outcomes have shown the importance of CBS testing engineering. At last, gaps between the conceptual approach and practical implementation will be solved.

The PA will consist of two phases i.e. requirements analysis and TCP. The major focus of PA will be to provide a solution of problems which arise in the maintenance phase of CBS development after changes by adding, removing and modifying components. Therefore, first, will be identified those components which will modify in all versions of development. Then extract their test cases to select for

execution to check CBS reliability. These selected test cases will be prioritized for higher faults detection rate on the base of different criteria i.e. frequently changed test cases, and faults history. This will improve CBS reliability and productivity within limited resource after changes.

In PA, customer/user initiated changes in components and then team identify changeable features of CBS then convert to feature model to identify commonalities and variabilities, if any ambiguity or incompleteness in required change then refine from customer otherwise implemented features and maintain all version change in components information in repository with complete change historical information. Then for TCP of component verification insert mutation faults and based on change history, extract test cases from test suites based on commonalities and variabilities for execution to identify relevant and earlier faults. For this test case, prioritization techniques will be adopted using frequently changed test cases criteria and their faults history. Then test cases of components executed and refined after faults removal then releases the product after reliability analysis.

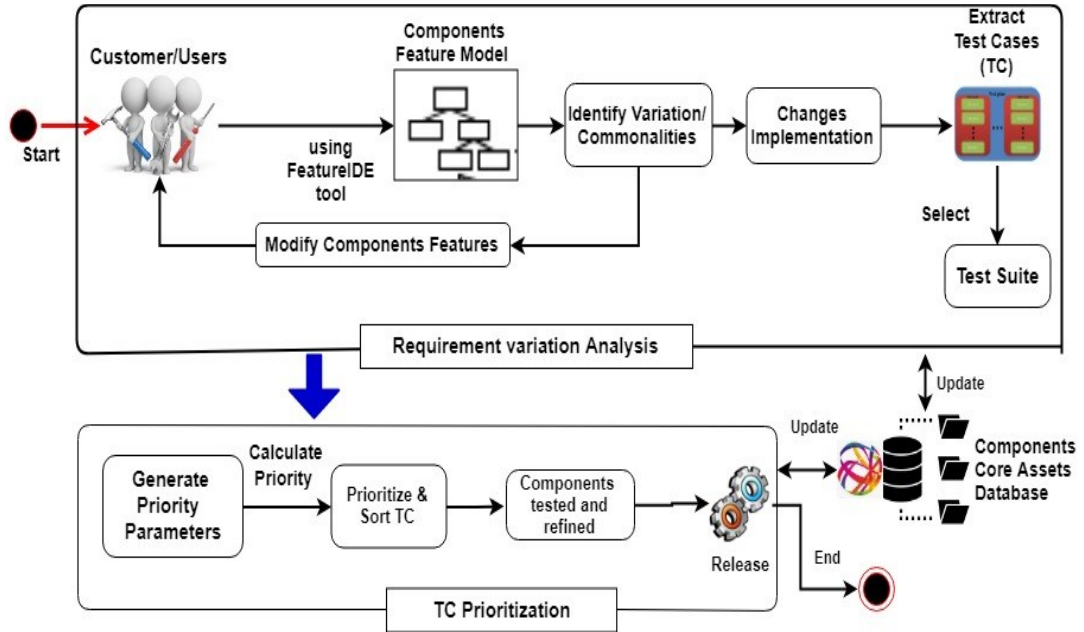


Figure 2. Possible proposed approach

IV. RESULTS AND WORK IN PROGRESS

For evaluation of proposed approach as an example; select open source two commercial web applications. First one is DASCP (a digital scan and archive designed for civil application web based written in ASP.Net. It includes Components 107, Functions 940, Files 201, LOC 35,122, Sessions 748, Faults 35, Versions 3, and Test Cases 95. Second example is online s and archive designed for civil application web based written in ASP.Net and used Microsoft SQL server. It includes Components 1900, Functions 940, Files 1501, LOC 236,152, Sessions 1480, Faults 75, Versions 15, and Test Cases 1925.

For components variation (CV) and commonalities (CC) analysis generates feature model. Then after implementation of changes for verification of changes extracted test suites and selected test cases (TC) based on CV and CC.

TABLE I. TEST CASES PRIORITY

TC	CF	PFR	NF	New TC
TC-1	1	0.45	0.45	TC-8
TC-10	0.76	0.20	0.76	TC-21
TC-17	0.37	0	0.37	TC-10
TC-21	0.80	0.25	0.80	TC-13
TC-2	0.47	0	0.47	TC-2
TC-8	0.82	0.30	0.82	TC-1
TC-9	0.67	0.35	0.35	TC-15
TC-3	0.33	0.10	0.10	TC-17
TC-12	0.33	0.15	0.15	TC-9
TC-13	0.57	0.27	0.57	TC-12
TC-15	1	0.40	0.40	TC-3

Then defined priority criteria based on frequently change test cases history and for tie cases (TC have same change frequency) defined second criteria for priority i.e. previous faults rate (PFR). In table 1 described sequence of prioritized TC for test execution with change frequency (CF) and PFR for similar TC frequency then new frequency (NF) calculated. New TC sorted from highest to lowest NF values.

TABLE II. FAULTS DETECTION RATES OF DASCP

TC	PA	TC	PFB
TC-8	100%	TC-16	40%
TC-21	80%	TC-19	20%
TC-10	70%	TC-20	0
TC-13	40%	TC-17	20%
TC-2	60%	TC-16	0
TC-1	50%	TC-2	10%
TC-15	40%	TC-14	10%
TC-17	30%	TC-4	40%
TC-9	20%	TC-22	20%

To verify that new TC of PA detected maximum faults in first execution and for comparative analysis of faults use previous faults (PFB) method for prioritization as described in Table 2 of DASCP. Whereas, in Table 3 described faults rate of Online Shopping application which identified after the execution of both approaches.

TABLE III. FAULTS DETECTION RATES OF ONLINE SHOPPING

TC	PA	TC	PFB
TC-18	96%	TC-28	46%
TC-23	87%	TC-60	25%
TC-1	72%	TC-22	20%
TC-3	45%	TC-31	30%
TC-8	50%	TC-46	15%
TC-1	50%	TC-10	10%
TC-25	48%	TC-14	18%
TC-70	35%	TC-4	35%
TC-33	23%	TC-22	38%

In Figure 3 described the companion of PA faults detection rate with PFB rate which near to 100 percent in PA case and less than 20 percent in case of PFB using DASCP dataset. In PFB we prioritized TC based on previous faults detection rate. While in Figure 4 depicted the comparison of rate after the implementation of PA (i.e. greater than 50 percent) and PFB on Online Shopping application (i.e. lessee than 50 percent).

Hence, x-axis in both Figures 3-4 explained number of test cases executed after prioritization with highest priority while y-axis explained percentage of rate of faults detected after every test case execution in both approaches i.e. PA and PFB using both datasets. Therefore, in both comparison PA significantly detected maximum faults as compared to PFB without deleting and reducing size of test cases.

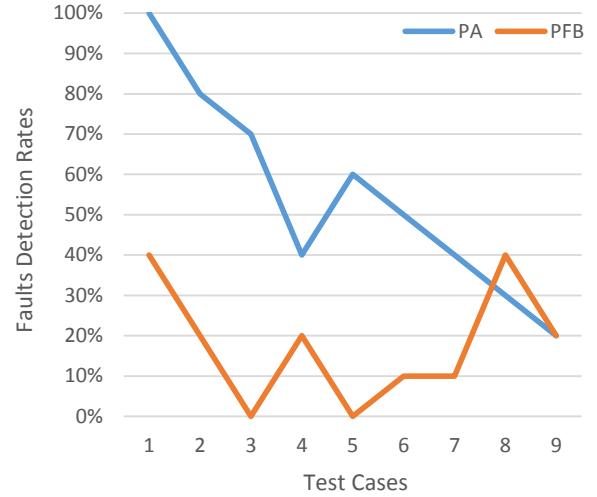


Figure 3. Comparison of DASCP Rates

From comparative analysis it derived that PA identified maximum and highest detection rate as compared to PFB method. The results described that PA reduce irrelevancy and redundancy as compared to PFB. And outperformed also in case of dynamic testing of component based systems.

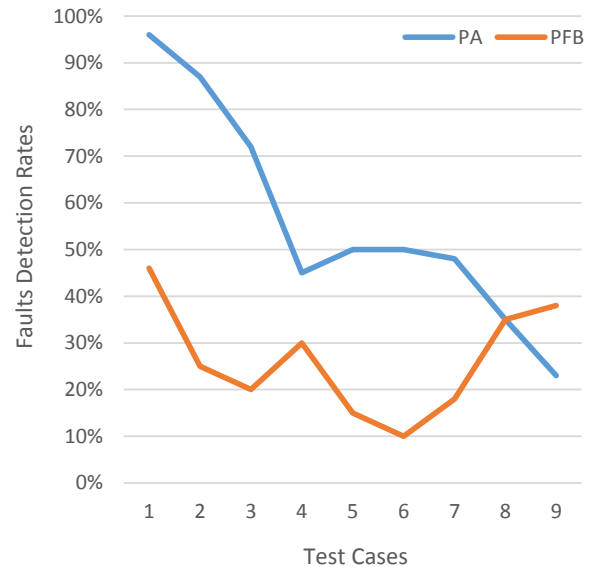


Figure 4. Comparison of Online Shopping Rates

Currently, implemented approach for regression testing in cyber physical system to ensure reliability and proof of concept for researchers and practitioners for requirements and test case prioritization with multi case studies. Then implement proposed approach as a prototype and verification of prototype conducted experiment.

V. COBCLUCION

In the research described the challenges of reliability analysis of modification in component based software development and proposed approach to facilitate CBS development process by removing identified limitations. Despite extensive research on CBS specification, analysis and

verification phases existed and has not impacted on industry practices due to availability in mis specification and difficult to manage commonalities and variabilities during Components integration. Thus, research bridges the gap between research and practice in the field of CBS regression testing.

REFERENCES

- [1] B. Hendradjaya, "A Proposal for New Software Testing Technique for Component Based Software System," *Int. J. Electr. Eng. Inform.*, vol. 10, no. 1, pp. 60–78, Mar. 2018.
- [2] R. M. Parizi, "Microservices as an Evolutionary Architecture of Component-Based Development: A Think-aloud Study," *ArXiv180511757 Cs*, May 2018.
- [3] S. Souto and M. d'Amorim, "Time-Space Efficient Regression Testing for Configurable Systems," *ArXiv170203457 Cs*, Feb. 2017.
- [4] S. C. Talukder and M. M. Rahman, "Customer requirements oriented component based software development life cycle model," in *2015 International Conference on Computers, Communications, and Systems (ICCCS)*, Kanyakumari, India, 2015, pp. 61–68.
- [5] N. Almasri, L. Tahat, and B. Korel, "Toward automatically quantifying the impact of a change in systems," *Softw. Qual. J.*, vol. 25, no. 3, pp. 601–640, Sep. 2017.
- [6] S. Y. Shin, S. Nejati, M. Sabetzadeh, L. C. Briand, and F. Zimmer, "Test case prioritization for acceptance testing of cyber physical systems: a multi-objective search-based approach," in *Proceedings of the 27th ACM SIGSOFT International Symposium on Software Testing and Analysis - ISSTA 2018*, Amsterdam, Netherlands, 2018, pp. 49–60.
- [7] M. Al-Hajjaji, T. Thüm, M. Lochau, J. Meinicke, and G. Saake, "Effective product-line testing using similarity-based product prioritization," *Softw. Syst. Model.*, vol. 18, no. 1, pp. 499–521, Feb. 2019.
- [8] A. Arrieta, S. Wang, G. Sagardui, and L. Etxeberria, "Search-Based test case prioritization for simulation-Based testing of cyber-Physical system product lines," *J. Syst. Softw.*, vol. 149, pp. 1–34, Mar. 2019.
- [9] S. Ghai and S. Kaur, "A Hill-Climbing Approach for Test Case Prioritization," *Int. J. Softw. Eng. Its Appl.*, vol. 11, no. 3, pp. 13–20, Mar. 2017.
- [10] M. Abu Hasan, M. Abdur Rahman, and M. Saeed Siddik, "Test Case Prioritization Based on Dissimilarity Clustering Using Historical Data Analysis," in *Information, Communication and Computing Technology*, vol. 750, S. Kaushik, D. Gupta, L. Kharb, and D. Chahal, Eds. Singapore: Springer Singapore, 2017, pp. 269–281.
- [11] P. Mahali and D. P. Mohapatra, "Model based test case prioritization using UML behavioural diagrams and association rule mining," *Int. J. Syst. Assur. Eng. Manag.*, vol. 9, no. 5, pp. 1063–1079, Oct. 2018.
- [12] H. Aman, T. Nakano, H. Ogasawara, and M. Kawahara, "A Topic Model and Test History-Based Test Case Recommendation Method for Regression Testing," in *2018 IEEE International Conference on Software Testing, Verification and Validation Workshops (ICSTW)*, Vasteras, 2018, pp. 392–397.
- [13] B. Miranda and A. Bertolino, "Scope-aided test prioritization, selection and minimization for software reuse," *J. Syst. Softw.*, vol. 131, pp. 528–549, Sep. 2017.
- [14] A. Shatnawi, A.-D. Seriai, H. Sahraoui, and Z. Alshara, "Reverse engineering reusable software components from object-oriented APIs," *J. Syst. Softw.*, vol. 131, pp. 442–460, Sep. 2017.
- [15] C. Magalhães, J. Andrade, L. Perrusi, and A. Mota, "Evaluating an Automatic Text-based Test Case Selection using a Non-Instrumented Code Coverage Analysis," in *Proceedings of the 2nd Brazilian Symposium on Systematic and Automated Software Testing - SAST*, Fortaleza, Brazil, 2017, pp. 1–9.
- [16] E. Kamburjan, S. Mitsch, M. Kettenbach, and R. Hähnle "Modeling and Verifying Cyber-Physical Systems with Hybrid Active Objects". *arXiv preprint arXiv:1906.05704*, 2019.
- [17] R. Eda, and H. Do. "An efficient regression testing approach for PHP Web applications using test selection and reusable constraints". *Software Quality Journal*, pp.1-35, 2019.
- [18] C. A. González, M. Varmazyar, S. Nejati, L. C. Briand and Y. Isasi. Enabling model testing of cyber-physical systems. In *Proceedings of the 21th ACM/IEEE International Conference on Model Driven Engineering Languages and Systems* (pp. 176-186). 2018 ACM.
- [19] R. Lachmann, "Machine Learning-Driven Test Case Prioritization Approaches for Black-Box Software Testing". *Proceeding-ettc2018*, pp.300-309.